```
`timescale 1ns / 1ps
//////////////////////////////////////////////////////////////////////////////////
// Company:
// Engineer:
//
// Create Date: 2026/10/15 14:16:28
// Design Name:
// Module Name: top_btn_debounce
// Project Name:
// Target Devices:
// Tool Versions:
// Description:
//
// Dependencies:
//
// Revision:
// Revision 0.01 - File Created
// Additional Comments:
//
//////////////////////////////////////////////////////////////////////////////////

module top_btn_debounce(
    input wire clk,
    input wire reset,
    input wire btnC,
    output [1:0] led
    );

    wire w_clean_btn;
    wire w_tick_1ms;

    button_debounce u_button_debounce(
        .clk(clk),
        .reset(reset),
        .btnC(btnC),
        .clean_btn(w_clean_btn)
    );

    led_toggle u_led_toggle(
```

```
        .clean_btn(w_clean_btn),
        .led(led[0])
    );

    tick_generator u_tick_generator(
        .clk(clk),
        .reset(reset),
        .tick_1ms(w_tick_1ms)
    );

    led_on_off_500ms u_led_on_off_500ms(
        .tick_1ms(w_tick_1ms),
        .led1(led[1])
    );

endmodule
```

```
`timescale 1ns / 1ps
//////////////////////////////////////////////////////////////////////////////////
// Company:
// Engineer:
//
// Create Date: 2026/10/15 15:55:16
// Design Name:
// Module Name: tick_generator
// Project Name:
// Target Devices:
// Tool Versions:
// Description:
//
// Dependencies:
//
// Revision:
// Revision 0.01 - File Created
// Additional Comments:
//
//////////////////////////////////////////////////////////////////////////////////
module tick_generator(
    input wire clk,
    input wire reset,
    output reg tick_1ms
    );
    // 100MHz 클록에서 1ms를 만들기 위한 카운터 (100,000 클록 주기)
    reg [$clog2(100000)-1:0] r_counter =0;
    always @(posedge clk or posedge reset) begin
        if (reset) begin
            r_counter <=0;
            tick_1ms  <=1'b0;
        end
        else begin
            if (r_counter == (100000 -1)) begin
                r_counter <=0;
                tick_1ms  <=1'b1;
            end
```

```
            else begin
                r_counter <=r_counter +1;
                tick_1ms  <=1'b0;
            end
        end
    end
endmodule
```

```
`timescale 1ns / 1ps
//////////////////////////////////////////////////////////////////////////////////
// Company:
// Engineer:
//
// Create Date: 2026/10/15 15:53:55
// Design Name:
// Module Name: led_on_off_500ms
// Project Name:
// Target Devices:
// Tool Versions:
// Description:
//
// Dependencies:
//
// Revision:
// Revision 0.01 - File Created
// Additional Comments:
//
//////////////////////////////////////////////////////////////////////////////////
module led_on_off_500ms(
    input  wire tick_1ms,
    output wire led1
    );
    reg [$clog2(500)-1:0] r_counter =0;
    reg r_led =1'b0;
    always @(posedge tick_1ms) begin
        if (r_counter ==500 -1) begin
            r_counter <=0;
            r_led     <=~r_led;
        end
        else begin
            r_counter <=r_counter +1;
        end
    end
    assign led1 =r_led;
endmodule
```